

3GPP TR 38.745 V20.0.0 (2026-03)

Technical Report

3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NG-RAN; Study on Artificial Intelligence (AI)/Machine Learning (ML) for NG-RAN Phase 3 (Release 20)



The present document has been developed within the 3rd Generation Partnership Project (3GPP™) and may be further elaborated for the purposes of 3GPP.
The present document has not been subject to any approval process by the 3GPP Organizational Partners and shall not be implemented.
This Specification is provided for future development work within 3GPP only. The Organizational Partners accept no liability for any use of this Specification.
Specifications and Reports for implementation of the 3GPP™ system should be obtained via the 3GPP Organizational Partners' Publications Offices.

3GPP

Postal address

3GPP support office address

650 Route des Lucioles - Sophia Antipolis
Valbonne - FRANCE
Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Internet

<http://www.3gpp.org>

Copyright Notification

No part may be reproduced except as authorized by written permission.
The copyright and the foregoing restriction extend to reproduction in all media.

© 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
All rights reserved.

UMTS™ is a Trade Mark of ETSI registered for the benefit of its members

3GPP™ is a Trade Mark of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners

LTE™ is a Trade Mark of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners

GSM® and the GSM logo are registered and owned by the GSM Association

Contents

Foreword	4
1 Scope	4
2 References	4
3 Definitions of terms, symbols and abbreviations	4
3.1 Terms	4
3.2 Symbols	5
3.3 Abbreviations	5
4 Use cases and Solutions	5
4.1 Multi-hop UE trajectory	5
4.1.1 Use Case Description	5
4.1.2 Solutions and Standard Impacts	5
4.1.2.1 Locations for AI/ML Model Training and AI/ML Model Inference	5
4.1.2.2 Input data of multi-hop UE trajectory	5
4.1.2.3 Output data of multi-hop UE trajectory	6
4.1.2.4 Feedback of multi-hop UE trajectory	6
4.1.2.5 Standard impact	6
4.2 AI/ML assisted Intra-CU LTM	6
4.2.1 Use Case Description	6
4.2.2 Solutions and Standard Impacts	6
4.2.2.1 Locations for AI/ML Model Training and AI/ML Model Inference	6
4.2.2.2 Input data of AI/ML based Intra-CU LTM	6
4.2.2.3 Output data of AI/ML based Intra-CU LTM	7
4.2.2.4 Feedback of AI/ML based Intra-CU LTM	7
4.2.2.5 Potential standard impacts	7
4.3 Other handover enhancements – AI/ML assisted inter-CU LTM	7
4.3.1 Use Case Description	7
4.3.2 Solutions and Standard Impacts	7
5 Conclusion	8
Annex A: Change History	9

Foreword

This Technical Report has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

- x the first digit:
 - 1 presented to TSG for information;
 - 2 presented to TSG for approval;
 - 3 or greater indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

1 Scope

The present document provides the description and investigation of new AI/ML based use cases, i.e., multi-hop UE trajectory, AI/ML based intra-CU LTM, and other handover enhancements.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

- [1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".
- [2] 3GPP TS 38.300: "NR; NR and NG-RAN Overall Description".
- [3] 3GPP TS 38.401: "NG-RAN; Architecture description".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

<Void>

3.2 Symbols

For the purposes of the present document, the following symbols apply:

<Void>

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

<Void>

4 Use cases and Solutions

4.1 Multi-hop UE trajectory

4.1.1 Use Case Description

In Rel-18, the cell-based UE trajectory prediction and measured cell-based UE trajectory are limited to the first-hop target NG-RAN node.

Predicted multi-hop UE trajectory across gNBs consists of a list of cells belonging to gNBs where the UE is expected to connect and these cells are listed in chronological order.

Measured multi-hop UE trajectory across gNBs consists of a list of cells belonging to gNBs where the UE was connected and these cells are listed in chronological order.

4.1.2 Solutions and Standard Impacts

4.1.2.1 Locations for AI/ML Model Training and AI/ML Model Inference

The following solutions are considered for supporting multi-hop UE trajectory:

- AI/ML Model Training is located in the OAM and AI/ML Model Inference is located in the gNB.
- AI/ML Model Training and AI/ML Model Inference are both located in the gNB.

In case of CU-DU split architecture, the following solutions are possible:

- AI/ML Model Training is located in the OAM and AI/ML Model Inference is located in the gNB-CU.
- AI/ML Model Training and Model Inference are both located in the gNB-CU.

4.1.2.2 Input data of multi-hop UE trajectory

To predict the multi-hop UE trajectory, a gNB may need the following information as input data:

From the UE:

- UE measurement report related to serving cell and neighbouring cells, e.g., RSRP, RSRQ, SINR
- Mobility History Information

From the neighbouring RAN nodes:

- UE History Information

From the local node:

- Measured multi-hop UE trajectory

4.1.2.3 Output data of multi-hop UE trajectory

For multi-hop UE trajectory, the following information can be considered as output data:

- Predicted multi-hop UE trajectory, including a list of cells that UE is expected to connect to in chronological order and the associated expected time UE stays in the cell

4.1.2.4 Feedback of multi-hop UE trajectory

For multi-hop UE trajectory, the following information can be considered as feedback:

- Measured UE Trajectory collected at each visited gNB

4.1.2.5 Standard impact

Multi-hop UE trajectory prediction is transferred to the target NG-RAN nodes over the Xn interface via the Handover Preparation procedure. For the subsequent handovers, multi-hop predicted UE trajectory includes a list of cells where the UE is expected to connect to in chronological order.

For the subsequent hops, Data Collection Reporting Initiation procedure needs to be in place between an initial source gNB and a subsequent gNB to request the reporting of measured UE trajectory. A subsequent gNB receives, via the Handover Preparation procedure, information needed to collect and then provide the measured UE trajectory. A subsequent gNB visited by the UE directly transmits the collected measured UE trajectory back to the initial source gNB over the Xn interface via the Data Collection Reporting procedure.

4.2 AI/ML assisted Intra-CU LTM

4.2.1 Use Case Description

L1/L2 Triggered Mobility (LTM) is specified in TS 38.300 [2].

Intra-CU LTM is specified in TS38.401[3].

AI/ML can be used to optimise Intra-CU LTM procedures, e.g., to enhance Network and UE performance, optimize resource allocation and reduce mobility failures.

AI/ML assisted L3 and L1 measurements based intra-CU LTM with inference in gNB-CU are studied.

4.2.2 Solutions and Standard Impacts

4.2.2.1 Locations for AI/ML Model Training and AI/ML Model Inference

For CU-DU split architecture, the following solutions are possible:

- AI/ML Model Training is located in the OAM and AI/ML Model Inference is located in the gNB-CU.
- AI/ML Model Training and Model Inference are both located in the gNB-CU.

4.2.2.2 Input data of AI/ML based Intra-CU LTM

For AI/ML optimization of intra-CU LTM the following information can be considered as input data:

From local node:

- L3 measurement results
- UE history information
- Measured/Predicted radio resource status per cell/SSB area
- Measured/Predicted cell-based UE trajectory
- Historical UE's candidate LTM cell and beam list(s)
- Measured TA values

From the UE:

- UE measurement reports
- Mobility history information

4.2.2.3 Output data of AI/ML based Intra-CU LTM

For AI/ML optimization of Intra-CU LTM the following information can be considered as output data:

- Predicted candidate cell(s) and beam(s) for LTM HO Preparation
- Predicted target cell(s) and beam(s) for cell switch command
- Predicted cell(s) and beam(s) for early UL synchronization
- Predicted TA value(s) for early UL synchronization
- Predicted timing to trigger cell switch for L3 measurement based LTM
- Predicted timing to trigger early UL synchronization for L3 measurement based LTM
- Predicted best beam of the 1st cell in the predicted cell-based UE trajectory
- Predicted TA validity time

NOTE: Applicability of the predicted TA validity time to a measured TA value and/or a predicted TA value will be assessed in the normative phase.

4.2.2.4 Feedback of AI/ML based Intra-CU LTM

The following data can be considered as feedback data for intra-CU LTM:

- LTM target cell and beam for cell switch
- Measured TA value(s)
- SON Reports (RLF, SHR, etc.)
- Timing of cell switch execution
- Timing of early UL synchronization

4.2.2.5 Potential standard impacts

F1 impacts are expected to support transferring of the information listed in clause 4.2.2.2, clause 4.2.2.3 and clause 4.2.2.4, if applicable.

4.3 Other handover enhancements – AI/ML assisted inter-CU LTM

4.3.1 Use Case Description

Inter-CU LTM is specified in TS 38.300 [2].

AI/ML may be used to optimise inter-CU LTM procedures, e.g., for candidate cell selection.

4.3.2 Solutions and Standard Impacts

If applicable, refer to the Solutions and standards impacts for intra-CU LTM in clause 4.2.2 to support inter-CU LTM with standards impacts over Xn interface.

5 Conclusion

The following use cases are recommended by RAN3 to be specified in the Rel-20 normative phase:

- Multi-hop UE trajectory
- AI/ML assisted intra-CU LTM
- AI/ML assisted inter-CU LTM

Recommended solutions and standard impacts for each use case are based on section 4.1.2, section 4.2.2, and section 4.3.2.

For each use case above, it is recommended to take the corresponding section “Solutions and standard impacts” (section 4.1.2, section 4.2.2, section 4.3.2) as baseline during the normative work.

Annex A: Change History

Change history						
Date	Meeting	TDoc	CR	Rev	Cat	New version
2025-10	RAN3#129-bis	R3-256546				Skeleton for TR38.745 v0.0.0
2025-10	RAN3#129-bis	R3-257331				Including text proposals R3-257329 and R3-257031 agreed in RAN3#129-bis.
2025-10	RAN3#129-bis	R3-257331				Update the version of the TR.
2025-11	RAN3#130	R3-258086				Submit to RAN3#130 for endorsement.
2025-11	RAN3#130	R3-258878				Including text proposals R3-258864, R3-258865 and R3-258866 agreed in RAM3#130.
2025-12	RAN#110	RP-253139				Submit to RAN#110 for approval.
2026-02	RAN3#131	R3-260027				Submit to RAN3#131 for endorsement.
2026-02	RAN3#131	R3-260821				Including text proposals R3-260670, R3-260724, R3-260806 and R3-260807 agreed in RAN3#131.
2026-03	RAN#111	RP-260263				Submit to RAN#111 for approval.
2026-03	RAN#111					TR Approved by RAN plenary